

EXPERIMENT

Aim of the Experiment: To determine the wavelength of laser light using diffraction pattern of single-slit and double-slit interference pattern.

Apparatus Required :

Laser, optical bench and optics kit containing double slits, single slits, diffraction grating and other accessories.

Experimental Procedure for Single Slit

1. Place laser and single slit on the optical bench and align on the optical bench. Such that light goes through the single slit.
2. Observe the diffraction pattern. If the pattern is too narrow and you cannot count the dark spots, use a single slit of different width.
3. Measure the distance between the single slit and the screen. Call this distance L .
4. Identify the central bright spot and compare its width and brightness to those of bright spots adjacent to it.
5. Identify the $m = 1$ dark spot and count dark spots outward. You should be able to count 6, 7 or more dark spots. Measure the distance between the outermost dark spots.
6. Calculate the angle θ_m .

7. Calculate the wavelength λ from the condition of destructive interference for the single slit

$$a \sin \theta_m = m\lambda \quad (1)$$

Where a is the slit width.

8. Compare to the accepted value of the wavelength by calculating the percent difference.

9. Does this procedure give a better result than procedure (1)

10. Draw a diagram showing the pattern you observed. Label the outermost dark spots with their numbers. On this diagram, show the distances you measured and the angle you calculated.

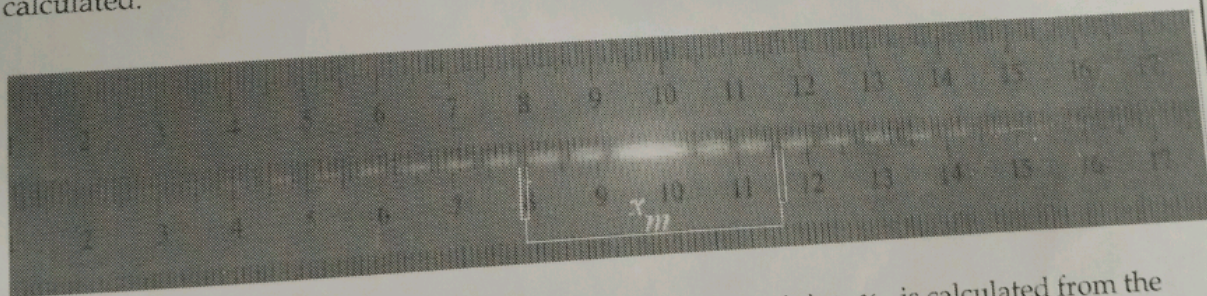


Figure 1. Single-slit diffraction pattern – distance between minima x_m is calculated from the Average minima position on either side of principal maxima.

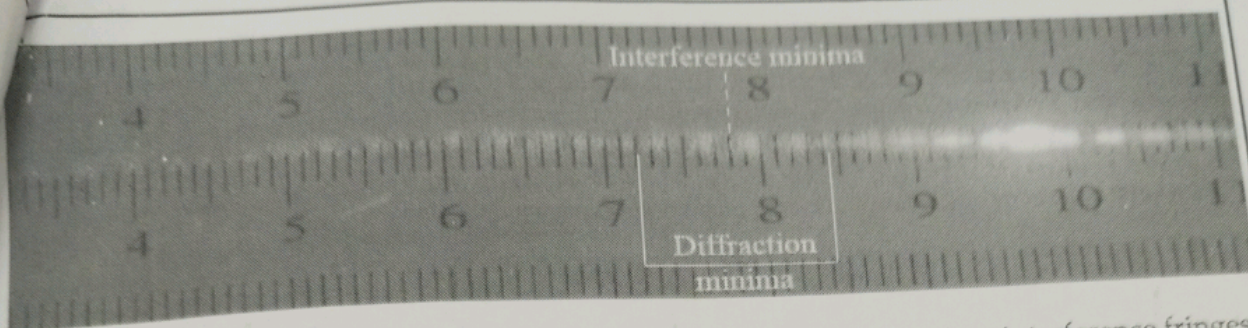
Observation Table

	Single Slit
Slits to screen distance $L(m)$	
Slit spacing, $d(m)$	
Fringe order, m	
Width of observed pattern (m)	
m th fringe angular position, θ_m	
Wavelength of laser light, λ	

Experimental Procedure for Double Slit

1. Place laser and double slit on the optical bench and align such that light goes through the double slit. Measure the distance between the double slit and the screen (should be around 2m). Call this distance L .
2. Observe the interference pattern on the screen. If the pattern is too narrow and the bright and dark fringes are too small to count, then use another double slit which will give you a wider pattern.
3. From the double slit plate, read the slit spacing. This is d .
4. Identify the $m = 0$ bright spot at the centre of the pattern. Count bright spots on either side of the centre bright spot until you get to a point where the bright spot looks like it is missing. You should be able to count 6, 7 or more bright spots. Measure the distance between the centres of the outermost bright spots.
5. Calculate the angle θ_m .
6. Calculate the wavelength λ from the constructive interference condition

$$d \sin \theta_m = m\lambda$$
7. Compare to the accepted value of the wavelength of the laser used by finding the percent difference.
8. Draw the pattern you observed as a series of bright and dark spots. Label the centre bright and the outermost bright spots with their numbers. Also on this diagram show the distances you measured and the angle you calculated.



Double-slit interference pattern – each diffraction maxima is broken up into interference fringes. The minima positions x_p (interference) and x_m (diffraction) are read off directly without averaging.

Observation Table

	Double Slit
Slits to screen distance $L(m)$	
Slit spacing, $d(m)$	
Fringe order, m	
Width of observed pattern (m)	
m th fringe angular position, θ_m	
Wavelength of laser light, λ	